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The EUV Snapshot Imaging Spectrograph

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1. INTRODUCTION

The light emitted by the solar transition region (TR) and corona varies significantly as a function of position, wavelength, and time. When viewed from Earth, the spectral radiance from the Sun can be written as: $I(x, y, \lambda, t)$, where x and y are the helioprojective Cartesian coordinates (W. T. Thompson 2006), λ is wavelength, and t is time. The ideal solar imaging spectrograph would capture $I(x, y, \lambda, t)$ with high resolution in x, y, λ, and t *and* over a wide field of view (FOV), wavelength range, and time period. Of course, the temporal dimension is privileged, so we often reduce the problem to capturing a 3D spatial/spectral cube at a particular time t_0: $I(x, y, \lambda, t_0)$. Since we use 2D detectors, this means that we must find a way to flatten the 3D cube into two dimensions without losing information.

One obvious way to accomplish this is to multiplex one of the three remaining dimensions in time. Narrowband, tunable filters, such as the GREGOR Fabry–Pérot Interferometer (K. G. Puschmann et al. 2012) or CRISP at the Swedish Solar Telescope (G. B. Scharmer et al. 2008), multiplex the wavelength dimension in time, and can change the selected wavelength in 50 ms or less (G. B. Scharmer et al. 2008), but the technology does not exist to use this technique for wavelengths shorter than ∼150 nm (J.-P. Wuelser et al. 2000). The nearest extreme ultraviolet (EUV) equivalent is a multilayer-coated imager such as the Transition Region and Coronal Explorer (TRACE) (B. N. Handy et al. 1999) or the Atmospheric Imaging Assembly (AIA) (J. R. Lemen et al. 2012), which abandons the wavelength dimension entirely and integrates over a passband holding many emission lines. Two such passbands can be compared to infer a Doppler shift (T. Sakao et al. 1999), but weaker lines within them bound the velocity resolution near $\sim$1000 km s^{-1} (K. Kobayashi et al. 2000), coarser than the flows that characterize TR dynamics.

A spatial dimension can be multiplexed instead. An entrance slit admits a single column of the scene, so the detector records λ against y without ambiguity, and x is recovered by stepping the slit across the target. The Interface Region Imaging Spectrograph (IRIS) (B. De Pontieu et al. 2014) and the Spectral Imaging of the Coronal Environment (SPICE) aboard Solar Orbiter (SPICE Consortium et al. 2020) work this way, and the spectra they return are the most faithful of any technique discussed here. What is sacrificed is simultaneity: neighboring columns of the reconstructed cube are separated in time by the raster, and structure that evolves faster than the raster completes is smeared along x. The Multi-slit Solar Explorer (MUSE) (B. De Pontieu et al. 2020, 2022; M. C. M. Cheung et al. 2022) narrows this gap considerably by ruling 37 slits across the field and separating their overlapping spectra afterward, which shortens the raster of an active region to as little as 12 s.

A third strategy divides the detector rather than the exposure. An integral field spectrograph slices the field of view and reformats those slices so that each is dispersed onto its own region of the detector, so the entire cube arrives in one exposure, at the cost of trading field of view against wavelength range. The microlensed hyperspectral imager of M. van Noort et al. (2022) demonstrates the technique at visible wavelengths, and the Solar eruptioN Integral Field Spectrograph (SNIFS) (V. L. Herde et al. 2024), flown in 2025, demonstrates it in the far ultraviolet (FUV). Extending it to the EUV remains out of reach of available optics.

The last strategy is to allow the flattening to be ambiguous and to undo it afterward. A spectrograph with no entrance slit disperses the whole field at once,

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so every emission line forms an image of the Sun and those images superimpose on the detector. The resulting frame, an *overlappogram*, encodes x and λ along a single axis, and the two cannot be separated within one exposure. Overlappograms are as old as spaceborne solar spectroscopy, having been recorded by the Naval Research Laboratory's (NRL's) S082A spectroheliograph (R. Tousey et al. 1973, 1977), which yielded both a census of EUV transitions (U. Feldman et al. 1985) and, much later, flare line ratios (F. P. Keenan et al. 2006); the technique is in use today at soft X-ray wavelengths by the Marshall Grazing Incidence X-ray Spectrometer (MaGIXS) (S. L. Savage et al. 2023). What has changed since S082A is that the ambiguity has become tractable. Reconstructing the cube from superimposed images is a tomographic problem (A. C. Kak & M. Slaney 1988), in which each diffraction order views the cube from a different angle and the number of independent views limits how much of the line profile can be recovered (M. R. Descour et al. 1997). Computed tomography imaging spectrographs (CTISs) (T. Okamoto & I. Yamaguchi 1991; F. V. Bulygin et al. 1991; M. Descour & E. Dereniak 1995) carry this to its extreme, projecting as many as 25 orders onto a single detector, with computational cost rising accordingly (N. Hagen & E. L. Dereniak 2008; N. Hagen & M. W. Kudenov 2013). Fewer views can suffice when the scene is sparse: C. DeForest et al. (2004) synthesized a magnetogram from a single exposure using only two dispersed orders. Inversion methods for overlapping spectral images have advanced considerably since (A. R. Winebarger et al. 2019; J. M. Davila et al. 2019; U. Kamaci & F. Kamalabadi 2026).

The Multi-Order Solar EUV Spectrograph (MOSES) (J. L. Fox et al. 2010; J. L. Fox 2011) brought this strategy to the solar EUV. A single concave grating forms three images at once, the undispersed $m = 0$ order and the $m = \pm 1$ orders, on three detectors, while a multilayer coating narrows the passband to a few emission lines so that the cube to be recovered is sparse enough to invert. Two flights showed that the concept works. J. L. Fox et al. (2010) measured the Doppler shift and width of explosive event line profiles in the TR; T. Rust & C. C. Kankelborg (2019) resolved bimodal profiles in quiet Sun events; H. T. Courrier & C. C. Kankelborg (2018) recovered Doppler maps using local correlation tracking; and J. D. Parker & C. C. Kankelborg (2022) separated the contributions of individual spectral lines. The same flights exposed where the design binds. Only three views are available, one of which is undispersed and therefore contributes no spectral information, which caps the number of degrees of freedom that an inversion can recover; and because the dispersed orders lie in a plane, they fill the cylindrical volume of a sounding rocket payload poorly.

The EUV Snapshot Imaging Spectrograph (ESIS) was designed to relax both constraints. An array of gratings arranged about the optical axis re-images the prime focus of a single primary mirror, each dispersing at a different angle, so that every detector records a dispersed view and the views are distributed around the payload rather than confined to a plane. ESIS flew for the first time on 2019 September 30, and first results from that flight are reported by J. D. Parker et al. (2022). An earlier form of the design was described by H. T. Courrier (2020); the instrument that flew differs from it, most consequentially in the removal of the primary mask, which admits vignetting into the system. This paper describes the instrument as it was built and flown. In Section 2 we explain how experience with MOSES shaped the design of ESIS. Section 3 states the scientific objectives and the requirements they impose. Section 4 describes the instrument itself, and Section **??** the mission profile.

2. THE ESIS CONCEPT

A primary goal of the ESIS instrument is to improve upon the imaging spectroscopy demonstrated by MOSES. Therefore, the design of the new instrument draws heavily from experiences and lessons learned through two flights of MOSES. ESIS and MOSES are both CTIS instruments. As such, both produce overlappograms of a narrow portion of the solar spectrum, with the goal of enabling the reconstruction of a spectral line profile at every point in the field of view. The similarities end there, however, as the optical layout of ESIS differs significantly from that of MOSES. In this section, we detail some difficulties and limitations encountered with MOSES, then describe how the new design of ESIS addresses these issues.

2.1. *Limitations of the MOSES Design*

The MOSES design features a single concave diffraction grating forming images on three charge-coupled device (CCD) detectors (J. L. Fox et al. 2010) (Figure 1). The optical path is folded in half by a single flat secondary mirror (omitted in Figure 1). Provided that the three cameras are positioned correctly, this arrangement allows the entire telescope to be brought into focus using only the central (undispersed) order and a visible light source. Unfortunately this design uses volume inefficiently for two reasons. First, the lack of magnification by the secondary mirror limits the folded length of the entire telescope to be no less than half of the 5 m focal length of the grating (J. L. Fox et al. 2010;

J. L. Fox 2011). Second, the resolving power of the instrument is limited by the ratio of camera separation to pixel size. To achieve the maximum dispersion of 29 km pix^{-1} s^{-1} (J. L. Fox et al. 2010), the outboard orders are imaged as far apart as possible in the ~ 0.56 m diameter envelope of the rocket payload. This planar arrangement leaves much unused space in the cylindrical volume of the payload.

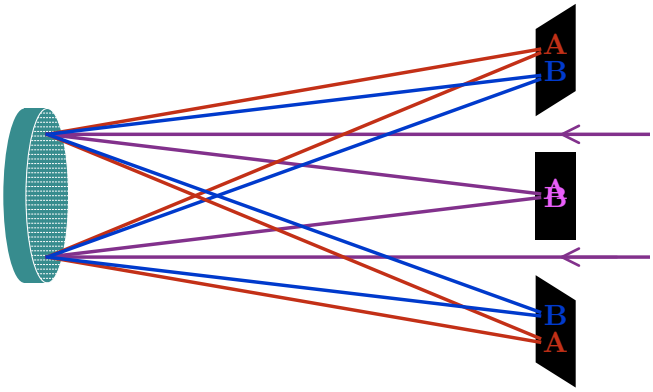


Figure 1. Schematic diagram of the MOSES instrument. Incident light from the right forms an undispersed image on the central $m = 0$ CCD. Dispersed images are formed on the outboard $m = \pm 1$ CCDs. The imaging of longer and shorter wavelengths is indicated by the use of red and blue letters respectively.

Furthermore, the monolithic secondary, though it confers the focus advantage noted above, does not allow efficient placement of the $m = \pm 1$ CCDs. For all practical purposes, the payload can only accommodate three diffraction orders ($m = -1, 0, +1$). Therefore, *MOSES can only collect, at most, three pieces of information at each point in the field of view.* From this, it is not reasonable to expect the reconstruction of more than three degrees of freedom for each spectral line, except in the case of very compact, isolated features such as those described by J. L. Fox et al. (2010) and T. Rust & C. C. Kankelborg (2019). Consequently, it is a reasonable approximation to say that MOSES is sensitive primarily to spectral line intensities, shifts, and widths (C. C. Kankelborg & R. J. Thomas 2001). With any tomographic apparatus, the degree of detail that can be resolved in the object depends critically on the number of viewing angles (A. C. Kak & M. Slaney 1988; M. R. Descour et al. 1997; N. Hagen & E. L. Dereniak 2008). So it is with the spectrum we observe with MOSES: more dispersed images are required to confer sensitivity to finer spectral details such as higher moments of the spectral line shape.

A related issue stems from the use of a single dispersion plane. Since the solar corona and transition region are structured by magnetic fields, the scene tends to be dominated by field-aligned structures such as loops (R. Rosner et al. 1978; R. M. Bonnet et al. 1980). When the MOSES dispersion direction happens to be aligned nearly perpendicular to the magnetic field, filamentary structures on the transition region serve almost as spectrograph slits unto themselves. The estimation of Doppler shifts then becomes a simple act of triangulation, and broadenings are also readily diagnosed (J. L. Fox et al. 2010; H. T. Courrier & C. C. Kankelborg 2018). A double-peaked profile can also be observed with sufficiently isolated features (T. Rust & C. C. Kankelborg 2019). Unfortunately, solar magnetic fields in the transition region are quite complex and do not have a global preferred direction. In cases where the field is nearly parallel to the instrument dispersion, spectral shifts and broadenings are not readily apparent.

The single diffraction grating also leads to a compromise in the optical performance of the instrument. Since the MOSES grating forms images in three orders simultaneously, there are not enough degrees of freedom in the optical system to control aberrations in all three spectral orders. During the first mission, MOSES was flown with a small amount of defocus (T. Rust & C. C. Kankelborg 2019), which exacerbated the inter-order point-spread function (PSF) variation and caused the individual PSFs to span several pixels (T. Rust & C. C. Kankelborg 2019; S. Atwood & C. Kankelborg 2018). The outboard PSFs were elongated, with their major axes at different angles. This resulted in spurious spectral features that require additional consideration (S. Atwood & C. Kankelborg 2018) and further increase the complexity of the inversion process (T. Rust & C. C. Kankelborg 2019; H. T. Courrier & C. C. Kankelborg 2018).

As MOSES lacks a field stop, it is possible for the grating to image off-band radiation on the detector from solar features that lie outside the intended FOV. Although the intensity of this off-band radiation is relatively low, it is not negligible. J. D. Parker & C. C. Kankelborg (2022) compared synthetic MOSES images to the real data and found that approximately ten percent of the intensity in the zeroth order image originated from more than ten dim lines in the MOSES passband, most of them too faint to be visible in the dispersed images. This study revealed that an undispersed ($m = 0$) channel, although attractive due to its lack of spatial-spectral ambiguity, is of limited utility due to spectral contamination. Moreover, the FOV should be clearly defined, and the same, for each image, so that the spectral and spatial content of each image is unambiguous.

Finally, the exposure cadence of MOSES is hindered by a $\sim$6 s readout time for the CCDs (J. L. Fox 2011).

The observing interval for a solar sounding rocket flight is typically about five minutes. Consequently, every second of observing time is precious, both to achieve adequate exposure time and to catch the full development of dynamical phenomena. The MOSES observing duty cycle is ∼50 % since it is limited by the readout time of its CCDs. Thus, valuable observing time is lost. The readout data gap compelled us to develop a sequence with exposures ranging from 0.25-24 s, a careful tradeoff between deep and fast exposures.

In summary, our experience leads us to conclude that the MOSES design has the following primary limitations:

1. inefficient use of volume
2. dispersion constrained by payload dimensions
3. too few dispersed images (orders)
4. single dispersion plane
5. lack of aberration control
6. poorly-defined and wavelength dependent FOV
7. spectral contamination
8. low duty cycle

In designing ESIS, we have sought to improve upon each of these points.

2.2. *ESIS Features*

The layout of ESIS (Figure 2) is a modified form of Gregorian telescope. Incoming light is brought to focus at an octagonal field stop by a parabolic primary mirror. In the ESIS layout, the secondary mirror of a typical Gregorian telescope is replaced by a segmented, octagonal array of diffraction gratings. From the field stop, the gratings re-image to CCD detectors arranged radially around the primary mirror. The gratings are blazed for first order, so that each CCD is fed by a single corresponding grating, and all the gratings are identical in design. The features of this new layout address all of the limitations described in Section 2.1, and are summarized here.

Replacing the secondary mirror with an array of concave diffraction gratings confers several advantages to ESIS over MOSES. First, the concavity of the gratings creates magnification in the ESIS optical system, which results in a shorter axial length than MOSES, without sacrificing spatial or spectral resolution. Second, the magnification and tilt of an individual grating controls the position of the dispersed image with respect to the optical axis, so that the spectral resolution is not constrained by the payload dimensions. Third, the radial symmetry of the design places the cameras closer together, resulting in a more compact instrument. Furthermore, by arranging the detectors around the optical axis, more dispersed grating orders can be populated; up to eight gratings can be arrayed around the ESIS primary mirror (up to six with the current optical table). This contrasts with the three image orders available in the planar symmetry of MOSES. Taken together, these three design features make ESIS more compact than MOSES (Section 2.1 item 1), improve spectral resolution (item 2) and allow the collection of more projections to better constrain the interpretation of the data (item 3).

The ESIS gratings are arranged in a segmented array, clocked in 45° increments, so that there are four distinct dispersion planes. This greatly aids in reconstructing spectral line profiles since the dispersion space of ESIS occupies a 3D volume rather than a 2D plane as with MOSES. For ESIS, there is always a dispersion plane within 22.5° of the normal to any loop-like feature in the solar atmosphere. As discussed in Section 2.1, a nearly perpendicular dispersion plane allows a filamentary structure to serve like a spectrographic slit, resulting in a clear presentation of the spectrum. This feature addresses Section 2.1 item 4.

Rather than forming images at three spectral orders from a single grating, each ESIS imaging channel has a dedicated grating. Aberrations are controlled by optimizing the grating design to form images in first order, over a narrow range of ray deviation angles. This design controls aberration well enough to allow pixel-limited imaging, avoiding the PSF mismatch problems inherent to the MOSES design (Section 2.1 item 5). In its flight configuration with gratings optimized around a 629.732 Å wavelength, the instrument cannot be aligned and focused in visible light like MOSES. Visible gratings and a special alignment transfer procedure (Section 4.8) are used to align and focus ESIS.

The ESIS design also includes an octagonal field stop placed at prime focus. This confers two advantages. First, the field stop fully defines the instrument FOV, so that ESIS is not susceptible to the spectral confusion observed in MOSES data (Section 2.1 item 6). Second, each spectral line image observed by ESIS is bordered by the outline of the field stop (e.g., Section **??**). This aids the inversion process since outside of this sharp edge the intensity is zero for any look angle through an ESIS data cube. The size and octagonal shape of the field stop are defined by the requirement that all CCDs must see the entire FOV from edge to edge, while leaving a small margin for alignment.

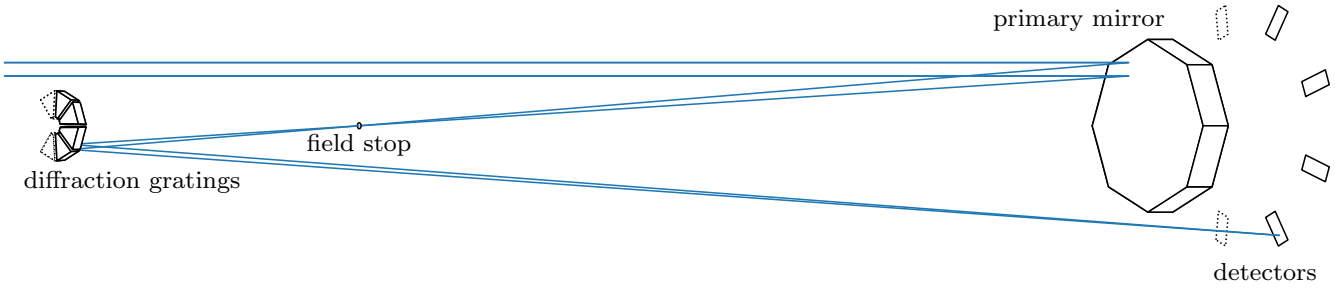


Figure 2. The ESIS optical layout. Dotted lines indicate the positions of unpopulated channels. The blue lines represent the path of O V through the system. The ESIS instrument is a pseudo-Gregorian design. The secondary mirror is replaced by a segmented array of concave diffraction gratings. The field stop at prime focus defines instrument spatial/spectral FOV. CCDs are arrayed around the primary mirror, each associated with a particular grating. Eight grating positions are available in principle; only six fit within the volume of the rocket payload. Four channels are populated for the first flight.

Lastly, in contrast to MOSES, ESIS employs frame transfer CCDs to make optimum use of our five minutes of observing time. The ESIS design is shutterless, so that each detector is always integrating. The result is a 100 % duty cycle. The lack of downtime for readout also allows ESIS to operate at a fixed, rapid cadence of 10 s. Longer integration times can be achieved for faint features by exposure stacking (Section 2.1 item 8).

In summary, the ESIS concept addresses all the limitations of the MOSES design enumerated in Section 2.1. The volume of the ESIS optical layout is smaller than MOSES by almost a factor of two, yet with a smaller PSF, improved spectral resolution, and faster exposure cadence. ESIS offers several features to improve the recovery of spectral information, including more channels, crossed dispersion planes, and a field stop.

3. SCIENCE OBJECTIVES

Early flights of MOSES demonstrated a working concept of simultaneous EUV imaging and spectroscopy. This concept adds a unique capability to the science that we can obtain from the EUV solar atmosphere. ESIS as designed improves upon the MOSES concept, as discussed in the previous section, and therefore improves our ability to accomplish our scientific objectives. In this section, we set forth the specific scientific objectives of the ESIS mission. It is from these objectives that we derived the quantitative science requirements (Section 3.3) that drove the ESIS design.

The ESIS mission was designed to achieve the following two overarching science goals: (1) observe magnetic reconnection in the TR, and (2) map the transfer of energy through the TR with emphasis on magnetohydrodynamic (MHD) waves. These objectives have significant overlap with the missions of IRIS (B. De Pontieu et al. 2014), the EUV Imaging Spectrograph (EIS) (J. L. Culhane et al. 2007) aboard Hinode, the EUV Normal-incidence Spectrometer (EUNIS) (J. W. Brosius et al. 2007, 2014), and a long history of FUV and EUV slit spectrographs. The ESIS instrument, however, can obtain both spatial and spectral information co-temporally. This allows us to resolve complicated morphologies of compact TR reconnection events (as was done with MOSES (J. L. Fox 2011; T. L. Rust 2017; H. T. Courrier & C. C. Kankelborg 2018)) and observe signatures of MHD waves over a large portion of the solar disk. Therefore, in support of goal 1, we will use ESIS to map flows as a function of time and space in multiple TR reconnection events. To achieve goal 2, we will cross-correlate the evolution at multiple temperatures in the TR to map the vertical transport of energy over a wide FOV.

3.1. *Magnetic Reconnection Events*

Magnetic reconnection describes the re-arrangement of the magnetic topology wherein magnetic energy is converted to kinetic energy resulting in the acceleration of plasma particles. Reconnection is implicated in many dynamic, high-energy solar events. Solar flares are a well-studied example (e.g., E. R. Priest & T. G. Forbes (2002) and the references therein); however, we have little hope of pointing in the right place at the right time to observe a significant flare event in a rocket flight lasting only five minutes. Instead, we will search for signatures of magnetic reconnection in TR spectral lines. A particular signature of reconnection in the TR is the explosive energy release by ubiquitous, small-scale events. These explosive events (EEs) are characterized as spatially compact (≈1.5 Mm length (K. P. Dere 1994)) line broadenings on the order of 100 km s^{-1} (K. P. Dere et al. 1991). They are observed across a range of TR emission lines that span temperatures of 20 000–250 000 K (C II–O V) (D. Moses et al. 1994). The typical lifetime of an EE is 60–90 s (D. Moses et al. 1994; K. P. Dere 1994; K. P. Dere et al. 1991). Due to their location near quiet Sun magnetic network elements, and the presence

of supersonic flows near the Alfvén speed, K. P. Dere et al. (1991) first suggested that EEs may result from the model of fast Petschek (H. E. Petschek 1964) reconnection.

The spectral line profile of EEs may indicate the type of reconnection that is occurring in the TR (e.g., T. L. Rust (2017)). For example, the Petschek model of reconnection predicts a 'bi-directional jet' line profile with highly Doppler-shifted wings, but little emission from the line core (D. E. Innes & G. Tóth 1999). D. E. Innes et al. (2015) developed a reconnection model resulting from a plasmoid instability (A. Bhattacharjee et al. 2009). In contrast to the bi-directional jet, this modeled line profile has bright core emission and broad wings. Both types of profile are seen in slit spectrograph data (e.g., D. E. Innes et al. (1997, 2015), and the references therein); however, MOSES observed EEs with more complicated morphologies than either of these two models suggest (J. L. Fox et al. 2010; T. L. Rust 2017). It is unclear whether the differing observations are a function of wavelength and temperature, a result of a limited number of observations, or a consequence of the morphology of the event being difficult to ascertain from slit spectrograph data.

ESIS will observe magnetic reconnection in the context of EEs, by extending the technique pioneered by MOSES to additional TR lines. Explosive events are well suited to sounding rocket observations; a significant portion of their temporal evolution can be captured in >150 s (e.g., the analysis by T. L. Rust (2017)) and they are sufficiently common to provide a statistically meaningful sample in a 5-minute rocket flight (e.g., K. P. Dere et al. (1989, 1991)). In similarity with MOSES, we sought a TR line for ESIS that is bright and well enough isolated from neighboring emission lines so as to be easily distinguished.

3.2. *Energy Transfer*

Tracking the mass and energy flow through the solar atmosphere is a long-standing goal in solar physics. Bulk mass flow is evidenced by Doppler shifts or skewness in spectral lines. However, the observed non-thermal broadening of TR spectral lines may result from a variety of physical processes, including MHD waves (B. De Pontieu et al. 2015, 2007), high-speed evaporative upflows (e.g., nanoflares, S. Patsourakos & J. A. Klimchuk (2006)), turbulence, and other sources (e.g., J. T. Mariska (1992)). This is a broad topic which ESIS can address in many ways. Here we will focus on a single application; ESIS will search for sources of Alfvén waves in the solar atmosphere by observing Doppler shifts and line broadening as the spectroscopic signature of these waves. The spectroscopic signature of Alfvén waves is most obvious near the limb of the solar disk, since the motion of the ions is transverse with respect to the propagation direction. Pointing near the limb may make it more difficult to observe distinct EEs, described in Section 3.1, since the average width of TR lines often increases near the limb of the Sun (T. R. Ayres 2021), thus decreasing the signal-to-noise ratio (SNR) of single events.

Alfvén waves in coronal holes are observed to carry an energy flux of 7×10^5 erg/cm^2/s, enough to energize the fast solar wind (M. Hahn et al. 2012; M. Hahn & D. W. Savin 2013). The source and frequency spectrum of these waves is unknown. Here, we hypothesize that MHD waves are similarly ubiquitous in quiet Sun and active regions, and play an important role in the energization of the quiescent corona.

The magnitude of non-thermal broadening of optically thin spectral lines is a direct measure of the wave amplitude (D. Banerjee et al. 2009; M. Hahn et al. 2012; M. Hahn & D. W. Savin 2013). We may estimate a lower limit on the non-thermal velocity to be observed as follows. We assume that the magnetic field is constant for small changes in scale height in the TR and that line-of-sight effects are negligible for observations sufficiently far from disk center. Since the solar wind is not accelerated to an appreciable fraction of the Alfvén wave velocity at altitudes $R \leq 1.15 R_\odot$ (S. R. Cranmer & A. A. van Ballegooijen 2005), the wave amplitude, v_{nt}, depends only weakly on electron density, n_e, so that $v_{nt} \propto n_e^{-1/4}$ (M. Hahn & D. W. Savin 2013; T. G. Moran 2001). Assuming pressure balance between the low corona and transition zone, we may infer non-thermal velocities in the TR by scaling according to the temperature drop, $v_{nt} \propto T^{1/4}$. The measured non-thermal velocity of 24 km s^{-1} for Si VIII (J. G. Doyle et al. 1998) (0.8 MK (T. G. Moran 2003)) near the limb should, neglecting damping, correspond to velocities of at least 18 km s^{-1} in the O V (0.25 MK) line. This is larger than the thermal width of 11 km s^{-1}.

More recently, A. K. Srivastava et al. (2017) observed torsional Alfvén waves with amplitude ∼20 km s^{-1} and period ∼30 s in the chromosphere. Modeling shows that these torsional waves can transfer a significant amount of energy to the corona (T. Kudoh & K. Shibata 1999). The torsional motion is observed as Doppler shifts when viewed from the side. The oscillation period is long enough to be well resolved but short enough to see ∼10 cycles in a single rocket flight. An ESIS-like instrument is therefore well suited to observations of torsional Alfvén wave propagation over multiple heights in the TR.

By mapping Doppler velocities over a wide field of view in the TR, ESIS can address questions about both the origin of waves and whether they are able to propagate upward into the corona. Independent of the two propagation modes discussed above, there is a range of possible sources for Alfvén (and other MHD) waves in the solar atmosphere. Three potential scenarios are: (1) Waves originate in the chromosphere or below and propagate through the TR at a spatially uniform intensity; (2) Intense sources are localized in the TR, but fill only a fraction of the surface; and (3) Weak sources are localized in the TR, but cover the surface densely enough to appear like the first case. The resulting non-thermal widths for localized sources are significantly higher than the $\sim$18 km s^{-1} lower limit derived above. The concentration of non-thermal energy observed by ESIS will serve as an indicator of source density. Comparison of Doppler maps captured at different temperatures by ESIS will indicate whether a uniform source density originates in the chromosphere or below (scenario 1) or is associated with spatially distributed TR phenomena (scenario 3) such as explosive events, or macrospicules. Comparison with a wider selection of ground- and space-based imagery will allow us to determine whether intense, localized sources (scenario 2) are associated with converging or emerging magnetic bipoles, type II spicules, spicule bushes, or other sources beneath the TR. For these comparisons, we need only to localize, rather than resolve, wave sources. A spatial resolution of $\sim$2 Mm is sufficient to localize sources associated with magnetic flux tubes that are rooted in photospheric intergranular network lanes (e.g., T. E. Berger et al. (1995)).

3.3. *Science Requirements*

ESIS will investigate two science targets: reconnection in explosive events and the transport of mass and energy through the transition region. The latter may take many forms, from MHD waves of various modes to EUV jets or macro-spicules. To fulfill these goals, ESIS will obtain simultaneous intensity, Doppler shift and line width images of the O V 629.732 Å line in the solar transition region (0.25 MK) at rapid cadence. The bright, optically thin O V emission line is well isolated except for the two coronal Mg X lines. These coronal lines can be viewed as contamination or as a bonus; we expect that with the four ESIS projections it is possible to separate the O V emission from that of Mg X. From the important temporal, spatial, and velocity scales referenced in Sections 3.1 and 3.2 we define the instrument requirements in Table 1 that are needed to meet our science goals.

4. THE ESIS INSTRUMENT

ESIS is a multi-projection slitless spectrograph that obtains line intensities, Doppler shifts, and widths in a single snapshot over a 2D FOV. Starting from the notional instrument described in Section 2, ESIS has been designed to ensure all of the science requirements set forth in Table 1 are met. The final design parameters are summarized in Table **??**.

A schematic diagram of a single ESIS channel is presented in Figure **??**a, while the mechanical features of the primary mirror and gratings are detailed in Figures **??**b and **??**c, respectively.

4.1. *Pointing System*

The imaging target was selected prior to launch, the morning of the day of flight. During flight, pointing was maintained by the Solar Pointing Attitude Rocket Control System (SPARCS) (Lockheed Missiles and Space Co. Report 1969). Images from Camera 1 were downlinked and displayed in real time on the SPARCS control system console at intervals of $\sim$16 s to verify pointing was maintained during flight.

4.2. *Mechanical*

ESIS and MOSES are mounted on opposite sides of a composite optical table structure originally developed for the Solar Plasma Diagnostics Experiment (SPDE) (M. E. Bruner 1995). The layered carbon fiber structure features a convenient, precisely coplanar array of threaded inserts with precision counterbores. The carbon fiber layup is designed to minimize the longitudinal coefficient of thermal expansion. The optical table is housed in two 0.56 m diameter skin sections, with a total length of 3 m. A ball joint and spindle assembly on one end and flexible metal aperture plate on the other hold the optical table in position inside the skin sections. The kinematic mounting system isolates the optical table from bending or twisting strain of the skins.

4.3. *Avionics*

The ESIS data acquisition and control system (DACS) is based on the designs used for both Chromospheric Lyman-alpha Spectro-polarimeter (CLASP) (R. Kano et al. 2012; K. Kobayashi et al. 2012; N. Narukage et al. 2016) and High-Resolution Coronal Imager (Hi-C) (K. Kobayashi et al. 2014; L. A. Rachmeler et al. 2019). The electronics are a combination of military off-the-shelf (MOTS) hardware and custom designed components. The DACS is a 6-slot, 3U, open VPX PCIe architecture conduction cooled system using an AiTech C873 single board computer. The data system also includes a MOTS PCIe switch card, Marshall Space Flight

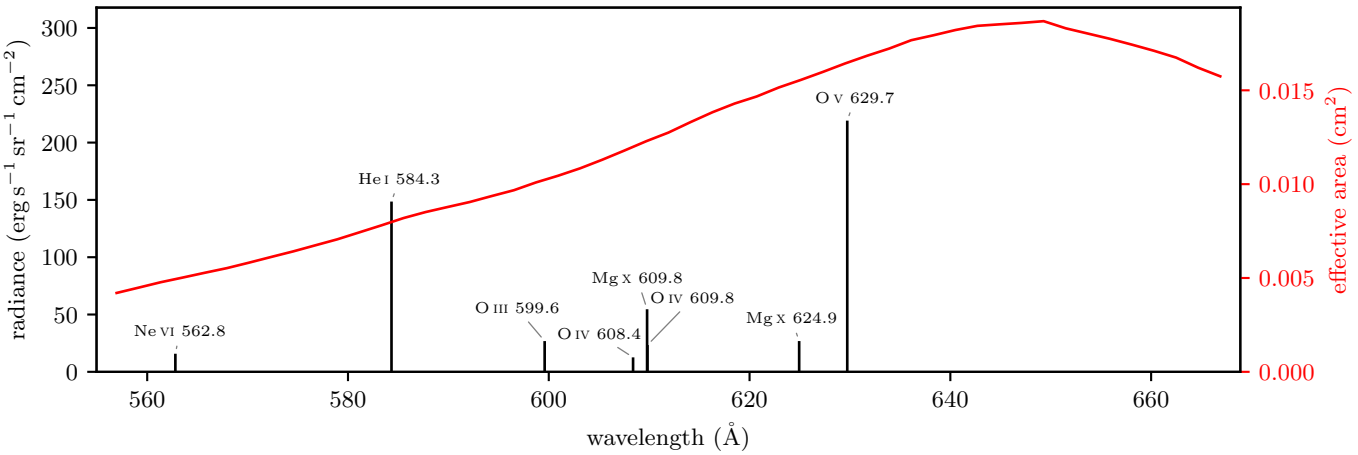


Figure 3. The eight brightest emission lines in the ESIS passband, and the effective area of a single channel beneath them. The lines are computed from version 11.0.2 of the CHIANTI atomic database, using the J. T. Schmelz et al. (2012) abundances of `sun_coronal_2012_schmelz_ext`, the quiet Sun differential emission measure (DEM) file `quiet_sun.dem`, and $n_e T = 1 \times 10^{15}$ K cm^{-3}.

Table 1. ESIS instrument requirements and capabilties. Note that MTF exceeds the Rayleigh criterion of 0.109.

Parameter	Requirement	Science Driver	Capabilities
Spectral line	O V 629.732 Å	EEs	O V, Mg X, He I, Figure 3
Spectral sampling	18 km s^{-1}	Broadening from MHD waves	17.5 km s^{-1} pix^{-1}, Table **??**
Spatial resolution	2.1″(1.5 Mm)	EEs	**??**, Table **??**
SNR	17.3 (CH)	MHD waves in CH	17.5 (15 × 10 s exp.), Table 3
Cadence	15 s	Torsional waves	10 s eff., Section 4.7
Observing time	150 s	EEs	306.8 s, Section **??**
FOV diameter	10′	Span QS, AR, and limb	11.5′, Table **??**

Center (MSFC) parallel interface card, and two MOTS Spacewire cards. A slot for an additional Spacewire card is included to accommodate two more cameras for the next ESIS flight. The C873 has a 2.4 GHz Intel i7 processor with 16 GB of memory. The operating temperature range for the data system is −40 °C to 85 °C. The operating system for the flight data system is Linux Fedora 23.

The DACS is responsible for several functions; it controls the ESIS experiment, responds to timers and uplinks, acquires and stores image data from the cameras, downlinks a subset of images through telemetry, and provides experiment health and status. The DACS is housed with the rest of the avionics (power supply, analog signal conditioning system) in a 0.56 m to 0.43 m transition section outside of the experiment section. This relaxes the thermal and cleanliness constraints placed on the avionics. Custom DC/DC converters are used for secondary voltages required by other electronic components. The use of custom designed converters allowed additional ripple filtering for low noise.

4.4. *Vignetting*

The original design of ESIS had no vignetting thanks to a stop placed at the primary mirror that was designed to perfectly fill the grating with the same amount of light for each point in the FOV. This is the ESIS design that was used for the optimization procedure of the grating parameters described in Section **??**, for example. All other results described in the paper use the fully-open system. Before flight, we decided to remove the primary aperture stop to increase the sensitivity of the instrument at the expense of introducing vignetting to the ESIS FOV. This was acceptable since the vignetting was found to be a simple linear field as shown in Figure 4, and could be removed in the post-processing phase.

4.5. *Distortion*

Imaging from the prime focus to the detector introduces distortion. Since the four ESIS channels are arranged in four different orientations, the distortion complicates the comparison of the channels, and we must correct for it. The distortion is due to two factors: first,

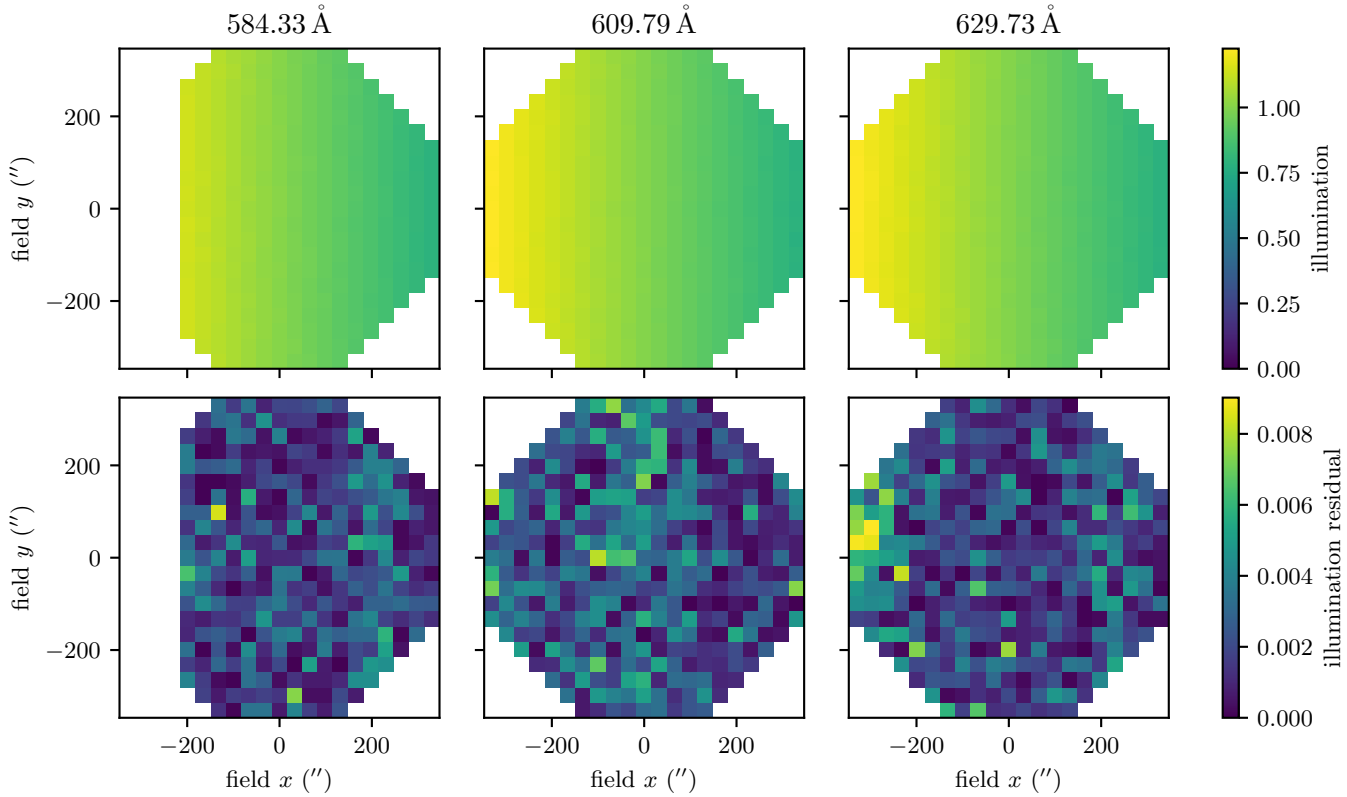


Figure 4. (Top) The relative illumination of a single ESIS channel as a function of position in the FOV, at each of the three target lines in the passband. The illumination is the fraction of the pupil which is unvignetted, normalized so that its average over the FOV is unity. (Bottom) The residual between that illumination and a linear model of it. Both rows are plotted only where the model was fit: the cells inside the FOV whose light lands on the detector. In the leftmost column, at He I, the FOV is cut off on the left by the edge of the detector.

the tilt of the detector as needed to maintain good focus over the FOV (L. Poletto & R. J. Thomas 2004); second, the anamorphic magnification of the grating (see F. Schweizer (1979)).

$$\begin{aligned}(x', y') = \mathbf{C} &+ \mathbf{C}_x x + \mathbf{C}_y y + \mathbf{C}_\lambda \lambda \\ &+ \mathbf{C}_{xx} x^2 + \mathbf{C}_{xy} xy + \mathbf{C}_{x\lambda} x\lambda \\ &+ \mathbf{C}_{yy} y^2 + \mathbf{C}_{y\lambda} y\lambda + \mathbf{C}_{\lambda\lambda} \lambda^2\end{aligned} \tag{1}$$

4.6. *Coatings and Filters*

The diffraction gratings are coated with a multilayer optimized for a center wavelength of O V 629.732 Å, developed by a collaboration between Reflective X-ray Optics LLC and Lawrence Berkeley National Laboratory (LBNL). In Figure 7, characterization of a single, randomly selected multilayer coated grating at LBNL shows that the grating reflectivity is constant over the instrument FOV in the $m = 1$ order while the $m = 0$ order is almost completely suppressed. Figure 8 shows a schematic of the coating that achieves peak reflectivity and selectivity in the $m = 0$ order using four layer pairs of silicon carbide (SiC) and magnesium (Mg). The

Table 2. Coefficients of the quadratic distortion model, Equation 1, of a single ESIS channel, fit at the three target lines in the passband. The field angles x and y are measured from the center of the FOV, and the wavelength λ from 607.95 Å, the mean of the three lines. The detector coordinates x' and y' are in pixels from the lower left corner of the light-sensitive area of the detector.

Coefficient		x'	y'
$\mathbf{C}$	(pix)	9.13×10^{2}	5.20×10^{2}
$\mathbf{C}_x$	(pix arcsec^{-1})	1.36	8.95×10^{-7}
$\mathbf{C}_y$	(pix arcsec^{-1})	-1.25×10^{-7}	1.30
$\mathbf{C}_\lambda$	(pix Å^{-1})	2.72×10^{1}	-7.79×10^{-6}
$\mathbf{C}_{xx}$	(pix arcsec^{-2})	6.92×10^{-6}	3.70×10^{-9}
$\mathbf{C}_{xy}$	(pix arcsec^{-2})	1.09×10^{-9}	4.40×10^{-6}
$\mathbf{C}_{x\lambda}$	(pix Å^{-1} arcsec^{-1})	2.52×10^{-4}	-4.10×10^{-8}
$\mathbf{C}_{yy}$	(pix arcsec^{-2})	2.07×10^{-6}	1.21×10^{-9}
$\mathbf{C}_{y\lambda}$	(pix Å^{-1} arcsec^{-1})	2.50×10^{-8}	9.37×10^{-5}
$\mathbf{C}_{\lambda\lambda}$	(pix Å^{-2})	2.45×10^{-3}	3.35×10^{-7}

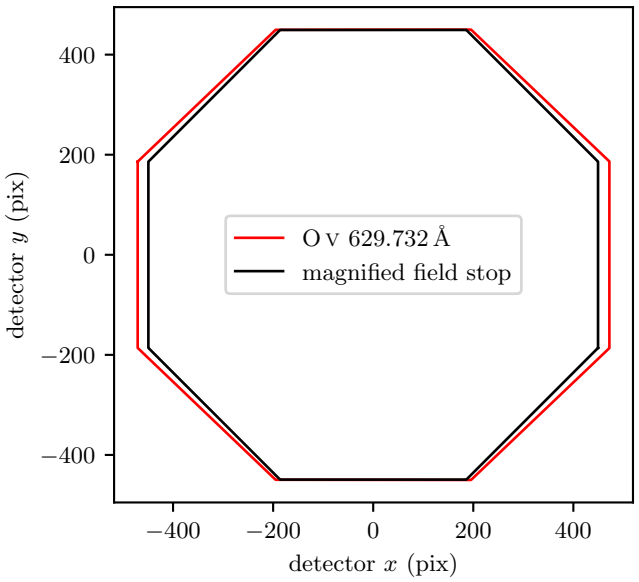


Figure 5. Plot of the magnified, undistorted field stop aperture vs. the distorted O v 629.732 Å image of the field stop aperture on the ESIS detector. The magnification factor used for the undistorted field stop aperture is the ratio of the grating exit arm to the grating entrance arm (4.01). The distorted image of the field stop aperture was calculated using the ESIS distortion model, described in Table 2.

aluminum (Al) layers are deposited adjacent to each Mg layer to mitigate corrosion.

The maximum reflectance for the coating alone in the nominal instrument passband is ∼35 % in the upper panel of Figure 9, measured from witness samples coated at the same time as the diffraction gratings. Combined with the predicted groove efficiency from Section **??** and, given the relatively shallow groove profile and near normal incidence angle, the total reflectivity in first order is ∼14 % at O v 629.732 Å. This is confirmed by the first order efficiency measured from a single ESIS grating in the lower panel of Figure 9.

Unlike EUV imagers (e.g., TRACE (B. N. Handy et al. 1999), AIA (J. R. Lemen et al. 2012), and the Hi-C (K. Kobayashi et al. 2014)) the ESIS passband is defined by a combination of the field stop and grating (Section **??**, Figure **??**) rather than multilayer coatings. The coating selectivity is therefore not critical in this respect, allowing the multilayer to be manipulated to suppress out-of-band bright, nearby emission lines. The lower panel of Figure 9 shows the peak reflectance of the grating multilayer is shifted slightly towards longer wavelengths to attenuate the He I 584.334 Å emission line, reducing the likelihood of detector saturation. A similar issue arises with the bright He II 303.785 Å line. Through careful design of the grating multilayer, the reflectivity at this wavelength is ∼2.5 % of that at O v 629.732 Å (upper panel of Figure 9). In combination with the primary mirror coating (described below) the rejection ratio at 303.785 Å is ∼37 dB. Thus, He II 303.785 Å emission is completely attenuated at the CCD.

The flight and spare primary mirrors were coated with the same Al/SiC/Mg multilayer. Corrosion of this multilayer rendered both mirrors unusable. The failed coating was stripped from primary mirror SN001. The mirror was then re-coated with a 5 nm thick layer of chromium (Cr) to improve adhesion followed by a 25 nm thick layer of SiC. The reflectance of this coating deposited on a silicon (Si) wafer witness sample appears in Figure 9. The spare primary mirror (SN002) retains the corroded Al/SiC/Mg multilayer.

The Si CCDs are sensitive to visible light as well as EUV. Visible solar radiation is much stronger than EUV, and visible stray light can survive multiple scatterings while retaining enough intensity to contaminate the EUV images. Luxél (F. R. Powell et al. 1990) Al filters 100 nm thick were used to shield each CCD from visible light. The Al film is supported by a 70 line per inch (lpi) nickel (Ni) mesh, with 82 % transmission. The theoretical filter transmission curve, modeled from CXRO data (B. L. Henke et al. 1993), is displayed in Figure 9. We conservatively estimate filter oxidation at the time of launch as a 4 nm thick layer of Al_2O_3.

An Al filter is positioned in front of the focal plane of each CCD by a filter tube, creating a light-tight box with a labyrinthine evacuation vent (e.g., Figure **??**). The placement of the filter relative to the CCD is optimized so that the filter mesh shadow is not visible. By modeling the filter mesh shadow, we find that a position far from the CCD (200 mm) and mesh grid clocking of 45° to the detector array reduces the shadow amplitude well below photon statistics. The MOSES instrument utilizes a similar design; no detectable signature of the filter mesh is found in data and inversion residuals from the 2006 MOSES flight.

To prevent oxidation, and to minimize the risk of tears, pinholes, and breakage from handling, the filters were stored in a nitrogen purged environment until after payload vibration testing.

4.7. *Sensitivity and Cadence*

Count rates for ESIS are estimated using the expected component throughput from Section 4.6 and the CCD quantum efficiency (QE) listed in Table **??**. Line intensities are derived from J. E. Vernazza & E. M. Reeves (1978) (V&R) and the Solar and Heliospheric Observatory (SOHO)/Coronal Diagnostic Spectrometer

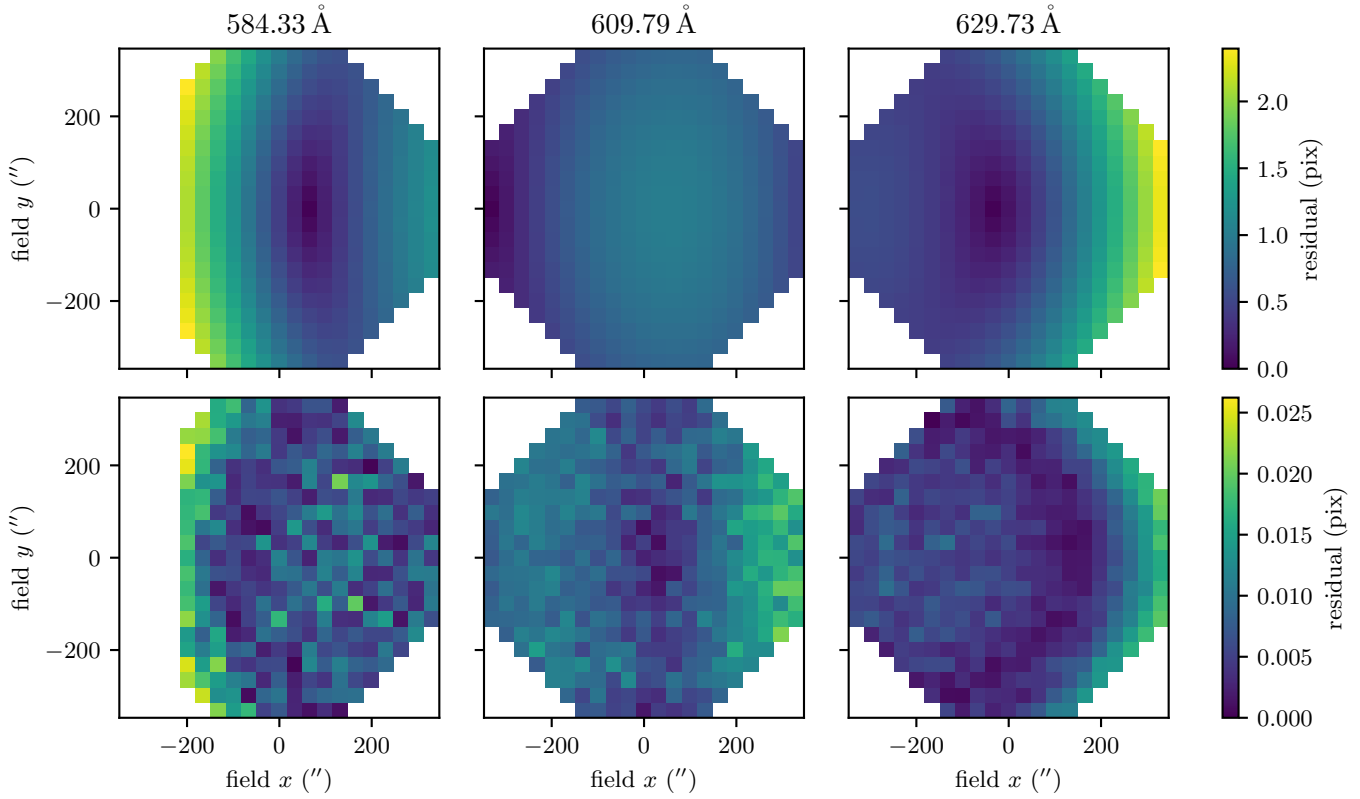


Figure 6. Magnitude of the residual between a linear distortion model and the raytrace model (top) and between a quadratic distortion model and the raytrace model (bottom). This figure demonstrates that a quadratic distortion model is sufficient to achieve sub-pixel accuracy.

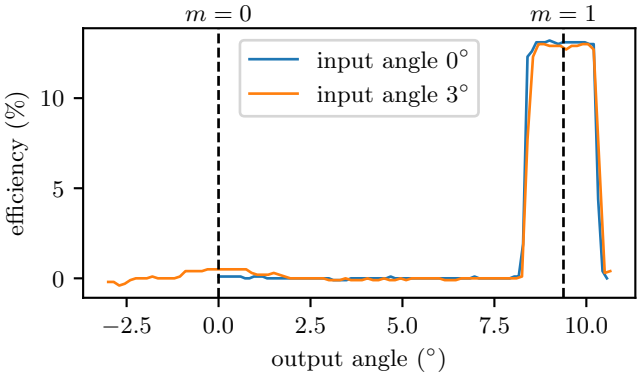


Figure 7. Measured efficiency at 63 nm of the Channel 2 grating as a function of reflection angle on 2018 January 21. Note flat response in first order over instrument FOV and suppression of zero order.

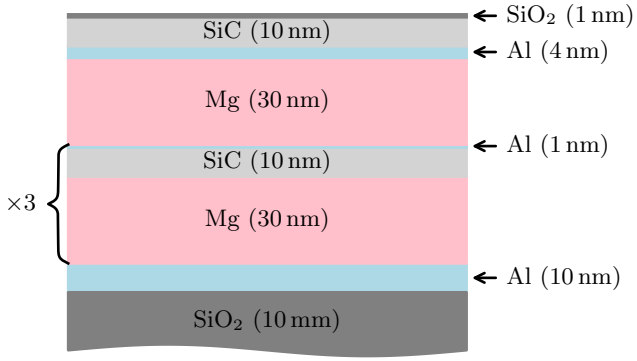


Figure 8. Schematic of the Al/SiC/Mg multilayer with $N = 4$ layers.

(CDS) (R. A. Harrison et al. 1995) data, and are given in a variety of solar contexts: quiet sun (QS), coronal holes (CHs), and active regions (ARs). The 100 % duty cycle of ESIS (Section **??**) gives us the flexibility to use the shortest exposures that are scientifically useful. So long as the shot noise dominates over read noise (which is true even for our coronal hole estimates at 10 s exposure length), we can stack exposures without a significant SNR penalty. Table 3 shows that ESIS is effectively shot noise limited with a 10 s exposure. The signal requirement in Table 1 is met by stacking exposures. Good quality images (∼300 counts) in active regions can be obtained by stacking 30 s worth of exposures. This cadence is sufficient to observe explosive events, but will not resolve torsional Alfvén waves described in Section 3. However, by stacking multiple 10 s exposures, sufficient

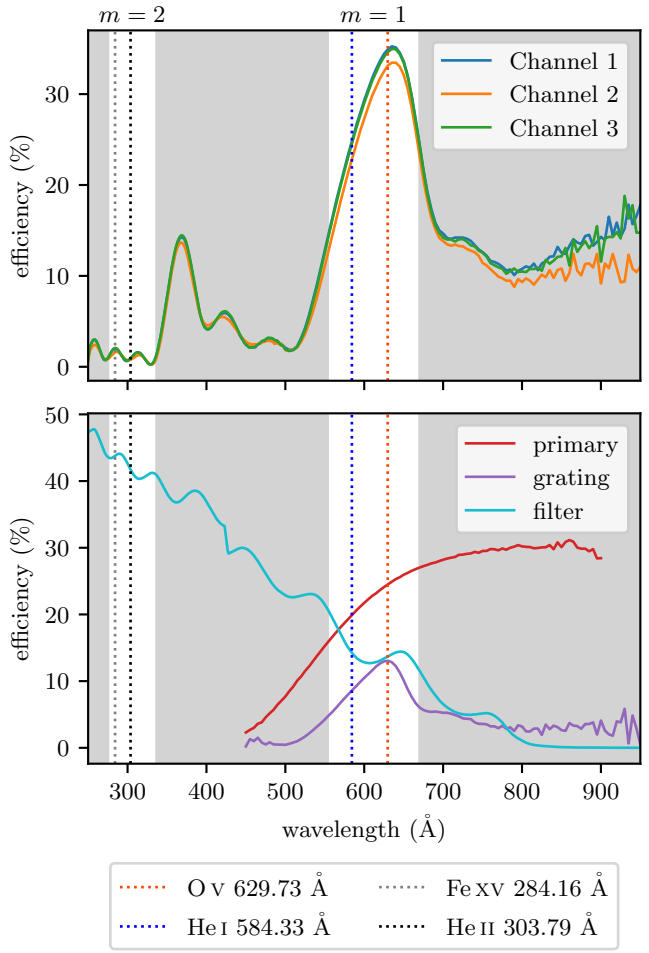


Figure 9. (Top) Measured reflectance for several multilayer coated witness samples at an incidence angle of 4° on 2018 January 21. The white regions indicate wavelengths that intercept the detector and the gray regions indicate wavelengths that miss the detector. Note the suppression of second order relative to the first order and the consistency of the coatings between each channel. The Channel 4 grating measurement is missing due to issues in the measurement apparatus. (Bottom) Comparison of the efficiency of the three main ESIS optical components: primary mirror, grating and filter. The primary mirror efficiency is based on measurements of a Si witness sample taken on 2018 May 10 at an angle of incidence of 4°. The grating efficiency is from a measurement of the Channel 2 grating taken on 2018 January 21 at an angle of incidence of 0°. The filter efficiency is a theoretical model that includes the filter mesh, 100 nm of Al and 4 nm of Al oxide.

SNR *and* temporal resolution of torsional Alfvén wave oscillations can be obtained.

4.8. *Alignment and Focus*

In the conceptual phase of ESIS, the decision was made to perform focus and alignment in visible light with a helium-neon (He-Ne) source. Certain difficulties are introduced by this choice, however, the benefits outweigh the operational complexity and equipment that would be required for focus in EUV. Moreover, a sounding rocket instrument requires robust, adjustment-free mounts to survive the launch environment. Such a design is not amenable to iterative adjustment in vacuum. The choice of alignment wavelength is arbitrary for most components; CCD response and multilayer coating reflectivity is sufficient across a wide band of visible wavelengths. The exceptions are the thin film filters (which were not installed until just before launch and have no effect on telescope alignment and focus) and the diffraction gratings. Visible light gratings were manufactured specifically for alignment and focus. These gratings are identical to the EUV flight version, but with a ruling pattern scaled to a 632.8 nm wavelength.

After alignment and focus had been obtained with the He-Ne source, the instrument was prepared for flight by replacing the visible gratings by the EUV flight versions. Each grating (EUV and alignment) is individually mounted to a backing plate using a bipod system similar to that of the primary mirror. The array of gratings on their backing plates are in turn mounted to a single part which we call the 'tuffet.' The backing plate is attached to the tuffet by bolts through spherical washers. With this mounting scheme, the gratings can be individually aligned in tip/tilt. The tuffet is attached to the secondary mirror mount structure (Figure 11). This enables the entire grating array to be replaced simply by removing/installing the tuffet, such as when switching between alignment and EUV gratings.

Properly positioning the gratings was expected to be the most difficult aspect of final telescope assembly. Ta-

Table 3. Estimated signal statistics per channel (in photon counts) for ESIS lines in CH, QS, and AR. Note that the SNR estimates are lower bounds since charge diffusion decreases the shot noise.

Source	V&R	V&R	V&R	CDS
Solar context	QS	CH	AR	AR
1 × 10 s exp.				
O V 629.732 Å	25	22	77	39
Mg X 624.941 Å	4	0	29	18
Total	29	22	106	57
Shot noise	5.4	4.7	10.3	7.5
Read noise	1.1	1.1	1.1	1.1
SNR	5.3	4.5	10.2	7.4
15 × 10 s exp.				
Total	435	326	1587	849
SNR	20.4	17.5	39.6	28.8

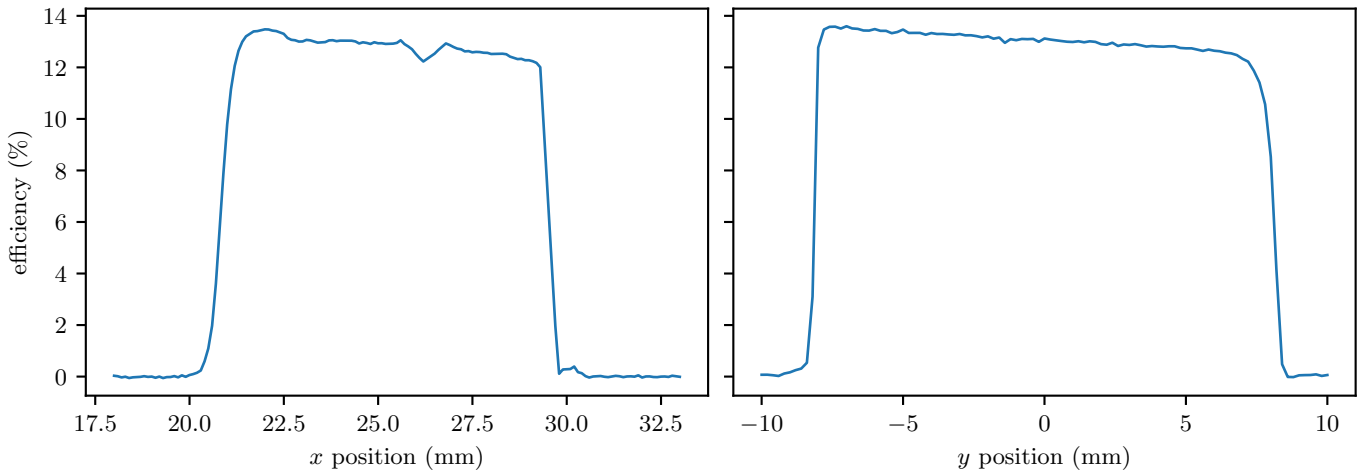


Figure 10. Channel 2 grating efficiency at 63 nm vs. position for two orthogonal slices across the optical surface on 2018 January 21.

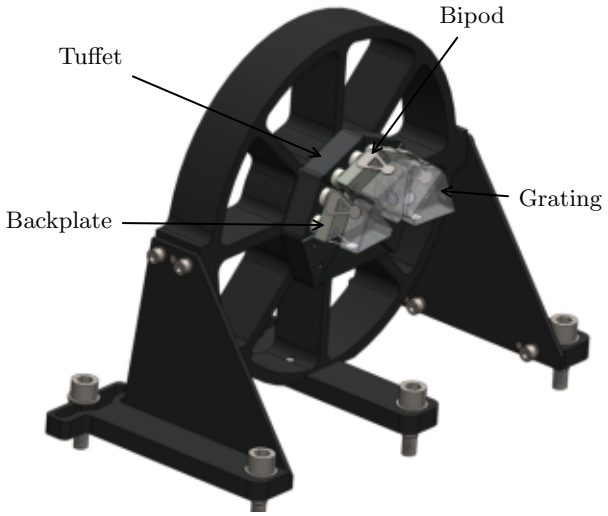


Figure 11. A solid model of the ESIS secondary mount. Each trapezoidal grating is supported and attached to its backplate by three bonded, titanium bipods. All mounted gratings sit on the "tuffet" and are mounted with three spherical washers. This allows independent tip/tilt adjustment of each grating, and for all of them to be removed/reinstalled without compromising alignment and focus.

ble **??** shows that the telescope is very sensitive to defocus. The depth of field is an order of magnitude smaller in EUV than in visible light. Moreover, the sensitivity to defocus at the gratings is $M^2+1 = 17$ times greater than at the detectors. Another sensitive aspect of the telescope is grating tip/tilt. A tolerance of $\sim \pm 0.34$ mrad is needed to ensure that the entire image lands on the active area of the CCD.

Once the visible gratings were aligned and focused, the challenge was to transfer this alignment to the EUV flight gratings. We performed this transfer using a 4D Phasecam 6000 interferometer. We aligned the interferometer to each visible grating such that the tip, tilt, and defocus aberrations were zero. Then, placing the corresponding flight grating in the visible grating's place, we shimmed the mounting screws of the flight grating to match the tip, tilt, and defocus of the alignment grating. Since the mounting interface for the tuffet is extremely repeatable by design we were able to swap tuffets and compare their alignment and focus to ensure everything was transferred correctly.

When transferring the focus and alignment two key details were considered. First, there is nothing about the tuffet that constrains the grating roll. Therefore, we needed to ensure each flight grating had the same roll as each visible grating. This was accomplished by using a He-Ne laser diverged through a cylindrical optic that illuminated each grating with a line perpendicular better make sure I know which direction is implied here to the grating blaze direction. The line of light was reflected back onto a ruled target that could be compared between gratings. Since our alignment gratings were ruled to image light of approximately an order of magnitude longer wavelength the laser and cylindrical optic were placed at the position of Littrow for the 10th order image in the visible grating, and the 1st order for the EUV gratings. Second, during testing we measured slight differences in radius of curvature between each grating. Therefore each flight grating was prescribed a specific amount of defocus to account for the difference in radius of curvature between each optic when transferring alignment and focus.

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